	Academic Semester	Fall 2026-2027	**Academic level**	1
	Course Code	MA104	**Exam**	Quiz 1 — Model E
	Course Title	Discrete Mathematics		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Salah Moawad , Dr. Tamer Seuodi		

Student Name: _____ **Student ID:** _____ **Section:** _____

*Answer **all** questions. Use T / F in truth tables and 1 / 0 in membership tables.*

Question 1 — Choose the correct answer and give the reason (5 points)

One point each: circle the letter of the correct answer **and** write a short reason in the box.

1. If p is **T** and q is **F**, which proposition is **false**?

A) $p \vee q$ **B)** $p \to q$
C) $q \to p$ **D)** $p \oplus q$

Reason:

2. Let $A = \{3, 5, 7, 9\}$ and $B = \{5, 9, 11\}$. Then $A - B =$

A) $\{11\}$ **B)** $\{3, 7\}$
C) $\{5, 9\}$ **D)** $\{3, 7, 11\}$

Reason:

3. $p \leftrightarrow q \equiv$?

A) $(p \to q) \vee (q \to p)$ **B)** $p \oplus q$
C) $(p \to q) \wedge (q \to p)$ **D)** $p \wedge q$

Reason:

4. Is $(p \to q) \wedge (\neg r)$ a **well-formed formula** (WFF)? If not, name the rule it breaks.

A) WFF **B)** Not a WFF

Reason:

5. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6\}$. Then $A \triangle B =$

A) $\{2, 4\}$ **B)** $\{1, 3, 5, 6\}$
C) $\{1, 3, 5\}$ **D)** $\{1, 2, 3, 4, 5, 6\}$

Reason:

Question 2 — Truth table (2.5 points)

(a) Complete the truth table of $\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$. [2 pts]

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$	$\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$
T	T						
T	F						
F	T						
F	F						

(b) Let P be the proposition in (a). Is $P \equiv \neg p \vee q$? Justify using your table. [½ pt]

Question 3 — Membership table (2.5 points)

(a) Complete the membership table for the identity $A \triangle B = (A \cup B) - A$. [2 pts]

A	B	$A \triangle B$	$A \cup B$	$(A \cup B) - A$
1	1			
1	0			
0	1			
0	0			

(b) Is the identity $A \triangle B = (A \cup B) - A$ **true** or **false**? Justify using your table. [½ pt]

— End of quiz · Good luck —